AutoOBE v2 — CO-PO Mapping & Attainment Platform

Engineering plan, corrected against the official NBA GAPC v4.0 SAR (January 2025)

- **Primary source:** NBA Self-Assessment Report Format, UG Engineering, Tier-II Institution Programs, GAPC v4.0, January 2025 — `nbaind.org/files/2-SAR-UG-EG-T2-14-1-2025_Format_20250115122255.pdf`
- **Target market:** AKTU-affiliated engineering colleges (Tier-II / affiliated, non-autonomous), then Anna University, VTU, RTU, MAKAUT.
- **Scope:** SAR Criteria 1, 3 and 8 in full, plus the outcome-mapping portions of Criterion 2 — approximately 440 of 1000 marks.

PART 0 — Stop. Your spec is built on a dead framework.

I pulled the official SAR. Six things in the plan you have are wrong in ways that would make the software's output unusable for an actual accreditation submission. Fix these before a line of code.

0.1 There are 11 Program Outcomes, not 12

NBA moved to **GAPC v4.0** (Graduate Attributes and Professional Competencies v4.0), aligned to the Washington Accord 2021 Review. Old PO6 (The Engineer and Society) and old PO7 (Environment and Sustainability) merged into a single new **PO6: The Engineer and the World**. Everything below shifted up one slot.

Old (GAPC v3)	New (GAPC v4.0)
PO1 Engineering Knowledge	PO1 Engineering Knowledge
PO2 Problem Analysis	PO2 Problem Analysis
PO3 Design/Development of Solutions	PO3 Design/Development of Solutions *(whole-life cost, net-zero carbon now explicit)*
PO4 Conduct Investigations	PO4 Conduct Investigations
PO5 Modern Tool Usage	**PO5 Engineering Tool Usage** *(renamed; "create, select and apply")*
PO6 The Engineer and Society	**PO6 The Engineer and the World** *(merged)*
PO7 Environment and Sustainability	**PO6** — this slot is gone
PO8 Ethics	PO7 Ethics *(diversity & inclusion added)*
PO9 Individual and Team Work	PO8 Individual and Collaborative Team Work
PO10 Communication	PO9 Communication
PO11 Project Management and Finance	PO10 Project Management and Finance
PO12 Life-Long Learning	PO11 Life-Long Learning

Effective January 2025 for Tier-I, January 2025 for Tier-II with a transition window that closed June 2025. Everything accrediting now is v4.0.

Engineering consequence: never hardcode PO count or PO codes. `program_outcomes` is a versioned reference table keyed by `(framework_version, po_code)`. You will ship a 12→11 migration tool on day one — every AKTU college has three years of legacy 12-PO matrices sitting in Excel, and NBA's position is that old batches need not be re-mapped but attainment **must be presented in the 11-PO structure**. That migration tool is a paid feature and a lead magnet.

Pull the exact statutory PO wording from Annexure II of the official PDF. Do not retype it, do not use a vendor blog's paraphrase. Your embeddings and your SAR export must both use byte-identical official text — an evaluator who spots reworded PO statements in your articulation matrix will discount the whole document.

0.2 WK1–WK9 is now a first-class entity, and nobody's software models it

GAPC v4.0 introduced the **Knowledge and Attitude Profile (WK1–WK9)** as Annexure I of the SAR, and the PO statements themselves now carry explicit WK references:

PO	WK linkage (per Annexure II)
PO1 Engineering Knowledge	WK1–WK4
PO2 Problem Analysis	WK1–WK4
PO3 Design/Development	WK5
PO4 Investigations	WK8
PO5 Engineering Tool Usage	WK2, WK6
PO6 The Engineer and the World	WK1, WK5, WK7
PO7 Ethics	WK9
PO8 Team Work	—
PO9 Communication	—
PO10 Project Management & Finance	—
PO11 Life-Long Learning	WK8

WK5 is the sustainability knowledge block (resource efficiency, whole-life cost, re-use, net-zero carbon). WK9 is ethics, inclusion and conduct. WK8 is research-literature engagement and critical thinking.

This is the single biggest product opportunity in the whole plan. The legacy tools were built for GAPC v3 and treat WK as a static annexure you paste in. Under v4.0, curriculum coverage of WK1–WK9 is what a peer team probes when they ask whether your PO claims are real. Build **CO → WK mapping as a first-class relation alongside CO → PO**, derive it partly from the PO linkage above and partly from syllabus unit content, and generate a **WK coverage heatmap across the whole curriculum**. No competitor ships this today. It is also cheap to build once you have the CO corpus.

0.3 Tier-II has 9 criteria, not 10, and they've been renamed around OBE

Vendor blogs are repeating "10 criteria." The official Tier-II format has nine, totalling 1000 marks:

#	Criterion	Marks	Product relevance
1	**Outcome-Based Curriculum**	120	**Core.** Vision/Mission/PEO mapping (40), curriculum structure (30), PO/PSO-course mapping (15), COs + Course Articulation Matrix (25), Program Articulation Matrix (10)
2	**Outcome-Based Teaching Learning**	120	Partial. Capstone quality (25), complex problems + SDGs (20), internships, seminars, MOOCs, industry partnership
3	**Outcome-Based Assessment**	120	**Core.** CIE/SEE quality (20), rubrics for labs/internships/projects (40), SDG evidence (10), CO attainment (25), PO/PSO attainment (25)
4	Students' Performance	120	Adjacent — enrolment ratio, success rate, API, placement, professional activities
5	Faculty Information	100	Out of scope (SFR, FQI, cadre proportion, retention)
6	Faculty Contributions	120	Out of scope
7	Facilities and Technical Support	100	Out of scope
8	**Continuous Improvement**	80	**Core.** Actions on CO attainment (20), actions on PO/PSO attainment (20), academic audit (15), academic performance improvement (15)
9	Student Support and Governance	120	Out of scope

Criteria 1 + 3 + 8 = 320 marks are almost entirely CO-PO data. Add the outcome-linked parts of Criterion 2 and you are addressing roughly **440 of 1000 marks**. That number is your entire sales pitch, and it's defensible because you can point at the official weightage table.

Scope your product to Criteria 1, 3, 8 and the outcome-mapping parts of 2. Do not build a faculty-records or facilities module. Those are ERP problems, they're where MasterSoft already wins, and they'll double your surface area for a quarter of the value.

0.4 Courses have up to ~6 COs, and K-levels are multi-valued

The SAR states the maximum number of outcomes for a course is expected to be **around 6**. Your schema assumes exactly 5. AKTU currently ships 5 per course, but colleges add a sixth for lab/project components and autonomous institutions vary. `CO1..CO5` as a fixed enum is a schema bug that will cost you a painful migration.

Worse: AKTU syllabus PDFs tag COs with **multiple** knowledge levels — real examples from the current AKTU B.Tech syllabus include `K2, K3`, `K3, K4`, `K1, K2`, and `K3, K6`. Your `blooms_level VARCHAR(20)` column cannot hold that. It must be `blooms_levels SMALLINT[]` or a join table, and every rule that reads "the CO's Bloom level" must be rewritten to read "the CO's *highest* Bloom level" with the full set retained for evidence.

0.5 PO attainment is a simple average across courses — not credit-weighted

The SAR is explicit: direct attainment of a PO/PSO is determined by taking the **average across all courses addressing that PO/PSO**. Table 3.8.1 is a courses × PO grid with a plain average row at the bottom.

I suggested credit-weighting in v1. That was wrong as a *default*. Credit-weighting is defensible and arguably better statistics, but it is a **deviation from the stated method** and must be declared in the program's OBE policy with justification. Ship NBA-literal as the default; offer credit-weighted as an opt-in with an auto-generated methodology note. Getting this backwards is exactly how a vendor produces numbers that don't reconcile with the evaluator's own spot-check.

0.6 "Target" has an official alternative definition you're not supporting

SAR 3.7.2: target may be stated as the percentage of students **getting more than class average marks**, or set by the program. A large fraction of colleges use the class-average definition precisely because it's self-normalising against a hard paper.

Your engine currently only supports a fixed threshold (60% of max marks). You need **both target modes**, selectable per course, with the mode printed in the report:

- `FIXED_THRESHOLD` — student passes if marks ≥ *k*% of question max
- `CLASS_AVERAGE` — student passes if marks ≥ class mean for that question

They produce materially different attainment values on the same data. Supporting only one means half your prospects can't use you without changing their published policy.

0.7 Two corrections to what I told you last time

- **AKTU end-semester question papers are not CO-tagged.** I checked actual AKTU B.Tech end-sem papers: the format is Section A / B / C with no CO or KL column. What *is* reliably tagged is (a) the **syllabus PDF**, which carries a `Course Outcome (CO) | Bloom's Knowledge Level (KL)` table for every course, and (b) **internal/sessional papers**, where most AKTU colleges use a template with CO and KL columns. The end-sem paper is still structurally useful — Sections B and C are one question per unit — so you can tag the *paper* unit-wise even though you can never get per-question student marks.
- **AKTU's own CO statements are frequently not NBA-compliant.** Verbatim from the current syllabus: "*Student should be aware of techniques used for classification and clustering*" and "*To provide hardware and software issues in modern distributed systems.*" Neither is measurable, neither is student-centric, neither uses a Bloom action verb. This is a genuine problem for every AKTU college and a product feature waiting to happen — see §3.6.

PART 1 — Product thesis

Three sentences:

> *Legacy OBE tools are data-entry surfaces with a report generator bolted on. The work they automate is the arithmetic, which was never the expensive part. The expensive part is getting defensible, correctly-tagged data in, and that is a document-parsing and inference problem.*

The automation contract. Publish this as a table on your landing page and hold yourself to it.

Work item	Legacy tools (per course/ semester)	AutoOBE	Human touchpoints
Define COs + K-levels	45–60 min	0 — from curriculum library	0
CO quality rewrite for NBA	not offered	0 — generated	1 click
CO→PO/PSO matrix + rationale	2–4 hrs	0 — generated with evidence	1 bulk approve
CO→WK mapping	not offered	0 — derived	0
Question paper Q→CO tagging	30 min per exam	0 — parsed from the paper	0
Mark entry	2–4 hrs per exam	2 min — upload or ERP sync	1 upload
CO attainment	1–2 hrs	event-driven, instant	0
PO/PSO attainment (program)	3–5 hrs	instant	0
Exit/alumni/employer surveys	days of chasing	auto-dispatch + reminders	0
CQI narrative (Criterion 8)	2 hrs	generated draft	1 click
Criteria 1/3/8 SAR assembly	2–4 weeks per program	minutes	review

Two faculty touchpoints per course per semester. Approve the matrix once (first time only), upload marks twice.

The design rule that makes this real: *no screen may ask a human to author anything.* Every screen arrives with a defensible default already computed. The human's job is to disagree, and disagreement must be one click with a reason field.

PART 2 — Domain model (v4.0-correct)

```
Framework (GAPC v3 | GAPC v4.0)          ← versioned, both must coexist
 ├── ProgramOutcome (po_code, statement_verbatim, wk_refs[])
 └── KnowledgeProfile (wk_code, statement_verbatim)

UniversityScheme (aktu, btech-cse, 2024-25)
 └── SchemeCourse (code, title, CI/LI/TW/SL hours, credits)
      └── SchemeCourseOutcome (co_code, statement, blooms_levels[])   ← array, not scalar

Institution (tenant)
 └── Program (dept, framework_version, psos[≤3])
      └── CourseOffering (scheme_course, academic_year, section, faculty)
           ├── CourseOutcome  (may override scheme CO)
           │    ├── CO_PO_Mapping   (weight 1|2|3, confidence, evidence[], approval)
           │    ├── CO_PSO_Mapping  (weight 1|2|3, …)
           │    └── CO_WK_Mapping   (derived + syllabus-grounded)
           └── Assessment (CIE | SEE | LAB | PROJECT | INTERNSHIP | SEMINAR)
                └── Question (q_no, co_id, blooms_level, max_marks)
                     └── StudentQuestionMark
```

Three entities that don't appear in your current schema and that the v4.0 SAR requires:

- **`Rubric`** / **`RubricCriterion`** / **`RubricScore`** — SAR 3.3, 3.4, 3.5 together are 40 marks and demand rubric-based assessment of labs, internships and projects, linked to POs/PSOs, with student-level evidence

retained in the course file. Project evaluation alone is 20 marks. Rubrics are not an optional nicety; they are the only accepted evidence path for PO8/PO9/PO10, which written exams cannot assess.

- **`SDGMapping`** — SAR 2.7 (20 marks) and 3.6 (10 marks) require evidence that student work addresses UN SDGs. A join from project/course to SDG codes with evidence artifacts. Trivial to build, 30 marks of SAR value, currently done by hand in Word.
- **`CurriculumHours`** — SAR 1.2.1 now demands CI / LI / (TW+SL) hours per semester under the National Credit Framework at 30 hours = 1 credit. AKTU publishes L-T-P, not NCrF hours. Build the L-T-P → NCrF converter once, for every course in the scheme. That's a one-afternoon feature that saves every college a week.

PART 3 — The auto-mapping engine

This is the IP. Everything else is CRUD and arithmetic that a competent team ships in six weeks.

Contract. Input: CO statement, `blooms_levels[]`, syllabus unit text, course type, assessment inventory. Output: for each of PO1–PO11 and each PSO, a weight in `{–,1,2,3}` plus an evidence bundle and a calibrated confidence.

3.1 Layer 1 — Bloom's verb prior

Curated verb lexicon → K-level → PO affinity prior. Deterministic, free, runs in microseconds.

```
apply / compute / solve / implement    K3 → PO1, PO2
design / develop / construct / create  K6 → PO3, PO5
analyse / compare / investigate        K4 → PO2, PO4
evaluate / justify / critique          K5 → PO4, PO6
present / document / report            any → PO9
collaborate / work in a team           any → PO8
```

Hard ceiling rule. A CO whose maximum K-level is K1 or K2 may not map to PO3, PO4 or PO5 at weight 3. Recall-level outcomes do not "design creative solutions." This one rule prevents the most common peer-team objection and costs nothing.

3.2 Layer 2 — Semantic similarity, calibrated

Embed the CO statement and each verbatim PO/PSO statement; cosine similarity; thresholds `τ1/τ2/τ3` fitted on a **hand-labelled gold set of ~500 CO-PO pairs** you build yourself from published SARs of already-accredited departments. Fit thresholds per discipline — CSE and Civil have completely different PO6 base rates.

Do not skip the gold set. Uncalibrated embedding similarity on this task produces confidently wrong 3s, and you will not know until an evaluator tells your customer.

3.3 Layer 3 — Assessment-evidence grounding

NBA's own definition of correlation strength is proportional to how much of the course actually engages that PO. So compute it from marks rather than guessing:

```
engagement(CO_i, PO_j) = Σ (marks of assessments tagged CO_i whose type channels to PO_j)
                         ÷ total course marks
```

Channel table: theory exam → PO1, PO2 · open-ended design problem → PO3 · lab experiment → PO4, PO5 · tool/simulation lab → PO5 · impact/safety/regulatory case study → PO6 · ethics module → PO7 · group project → PO8 · report + viva → PO9 · project with schedule/costing → PO10 · seminar/NPTEL/self-study → PO11.

This makes every weight traceable to a specific mark in a specific assessment. That traceability is the thing an evaluator actually wants and the thing no competitor produces.

3.4 Layer 4 — WK derivation

Two paths, combined: 1. **Inherited** — if CO maps to PO*j* at weight *w*, it inherits PO*j*'s declared WK refs at proportional strength. 2. **Grounded** — semantic match of syllabus unit content against the WK1–WK9 statements. A unit on lifecycle costing or energy efficiency hits WK5 regardless of which PO the CO was mapped to.

Roll up to a program-level **WK coverage heatmap**. Uncovered WKs are a curriculum gap you surface to the HOD — which is also SAR 1.2.3's "identified curricular gaps," worth marks directly.

3.5 Layer 5 — LLM arbitration, constrained

Only now call a model, and constrain it hard: give it the CO, the verbatim PO, the K-levels, the Layer 1–3 signals and the syllabus units; force a JSON schema `{po_code, weight, rationale_40_60_words, evidence_refs[]}`; temperature 0; all POs in one call per CO for internal consistency; two-pass self-consistency with disagreements flagged.

The model's job is to **arbitrate between the prior layers**, not to invent from scratch. Rationales must cite evidence — "*assessed via CT1 Q3 and CT2 Q5, 18% of internal marks, open-ended design questions*" — not produce prose.

Pin the model version per academic year and store it on every mapping row. When you re-run 2026's numbers in 2028 during a re-accreditation audit, they must come out identical.

3.6 Layer 5b — CO statement rewriter (do not skip this)

AKTU's published COs are frequently unusable as written. The rewriter takes "*Student should be aware of techniques used for classification and clustering*" and produces "*Apply supervised and unsupervised classification techniques to categorise data sets (K3)*" — preserving scope, adding an action verb, making it measurable.

Present as a diff: original on the left, proposed on the right, faculty accepts or edits. Retain **both** — the university-published CO for provenance, the department-adopted CO for attainment. This distinction matters: an affiliated college may legitimately sharpen a CO statement, but it must be able to show what it started from.

Colleges are currently paying consultants ₹50k–₹2L per department for exactly this. You're doing it in a click.

3.7 Layer 6 — Structural constraint solver

Raw model output over-maps. Every row passes hard constraints **before** persisting, and violations trigger a re-run, not a warning:

1. At most 6 POs per CO (accepted range 3–6).
2. At most two weight-3 entries per CO.
3. PO6, PO7, PO10 must be sparse across technical courses. If a Data Structures course maps every CO to PO6, reject.
4. Column mean ≤ 2.6 for any PO across the course's COs, unless that course is explicitly that PO's home course.
5. Every PSO must have ≥1 CO at weight ≥2 somewhere in the program, else the PSO is an orphan.
6. Program-wide: every PO must be addressed at weight ≥2 by ≥N courses. Orphan POs are an automatic finding, and this check feeds SAR Table 1.3.2 (PO → list of courses) directly.

Constraints are **code**, not prompt text. A prompt can be ignored; an assertion cannot.

3.8 Layer 7 — Confidence gating

```
confidence = w1·layer_agreement + w2·evidence_coverage + w3·llm_self_consistency
```

- `≥ 0.85` → auto-accepted, presented for bulk endorsement
- `0.60–0.85` → review queue, pre-filled, one-click accept
- `< 0.60` → mandatory decision, three competing suggestions shown side by side

Target ≥85% auto-accept. A full course matrix should be a 90-second review.

3.9 Layer 8 — Audit trail

Every mapping row stores engine version, model version, framework version, all layer scores, evidence refs, `generated_at`, `approved_by`, `approved_at`, plus full revision history. When a peer team asks "*who decided this was a 3, and on what basis?*", the answer is on screen in two seconds.

This is simultaneously your compliance story, your differentiator, and your switching cost. Three years in, a customer cannot leave without abandoning their evidence chain.

PART 4 — Ingestion

4.1 Curriculum corpus — build this first, and build it yourself

Batch-parse the AKTU syllabus PDF set into structured records: course code, title, L-T-P, credits, five units, CO statements, K-level arrays. Store as a versioned **global curriculum library** keyed by `(university, program, scheme_year)`. Colleges subscribe to a scheme; they never type a syllabus.

Multiple schemes must run concurrently — final-years on the 2021 scheme, first-years on 2024.

This is the moat. A competitor can copy the mapping engine in a quarter; copying 250 parsed and hand-verified courses per branch, with correct K-level arrays, is months of unglamorous work that nobody wants to do. Do it once, before launch.

4.2 Question paper parsing

Most internal papers are DOCX. **Parse the OOXML tables directly** — no OCR, near-100% accuracy, and the CO/KL columns most AKTU colleges already print give you Q→CO tagging for free.

```
QP upload
 → DOCX: direct XML table extraction | PDF: pdfplumber table extraction | scan: table-structure OCR
 → columns: Q.No | Question | Marks | CO | KL
 → validate: Σ marks == stated total; every CO exists; KL ≤ CO's declared max K-level
 → auto-create assessment + question rows
 → surface only mismatches
```

Ship a **QP quality audit** on top: K-level distribution, CO coverage balance, and a warning when >60% of the paper sits at K1/K2. Faculty get a one-page "this paper is 78% recall" report *before* it goes to print. It's genuinely useful, it improves the data feeding everything downstream, and it maps to SAR 3.1/3.2 (20 marks on assessment quality).

For the AKTU end-sem paper, tag **unit-wise** from Sections B/C structure. You'll never get per-question marks, but a unit-tagged SEE paper is required evidence for SAR 3.2 regardless.

4.3 Marks — the ROI ladder

1. **ERP connector** (read-only view or API, nightly sync). Zero faculty effort. Build for the two ERPs most common among your first ten customers, not "all ERPs."
2. **Excel/CSV with column auto-detection.** Do not ship a template. Detect roll numbers by regex, question columns by header pattern (`Q1a`, `1(a)`, `Q.1a`), infer max marks from the already-parsed QP. Preview the mapping, let the user correct once, then **remember that layout per faculty forever**.
3. **LMS API** (Google Classroom, Moodle) for assignments and quizzes.
4. **Printed award-sheet OCR** — table-structure OCR with per-cell confidence, low-confidence cells highlighted.
5. **Handwritten OCR** — last, always with a side-by-side diff.

Ordering matters. Handwriting OCR at 85–92% accuracy means every sheet needs verification, which is slower than typing. Leading with it is how OBE products die.

Validation on every ingest: marks ≤ max, roll numbers match enrolment, absentee detection, duplicate detection, outlier flag (zero on one question and full marks on the rest is a data-entry error, not a student).

4.4 University results

Parse the college's own downloaded AKTU result exports — subject-wise totals per roll number. Compute `% students ≥ target`. Never scrape the portal.

4.5 Indirect assessment

SAR 3.8.2 requires named surveys with PO-wise indirect attainment values. Automate end-to-end:

- Generate survey items **directly from CO statements** — "*Rate your ability to apply graph algorithms to routing problems*", Likert 1–5.
- Auto-dispatch by email / WhatsApp / SMS at semester end; three auto-reminders; auto-close.
- Course-exit, program-exit, alumni, employer — four survey types, each mapping to POs.
- Response-rate dashboard with a threshold warning. Evaluators question surveys below ~60% response, and a low-response survey used as 20% of PO attainment is a finding waiting to happen.

PART 5 — Attainment engine

Rule: no AI anywhere in this path. AI touches mapping suggestions and narrative drafts. Every number that appears in a SAR is produced by deterministic SQL.

5.1 CO attainment — internal (CIE)

```
target_marks(q) = max_marks(q) × target_pct              # FIXED_THRESHOLD mode
                | class_mean_marks(q)                    # CLASS_AVERAGE mode (SAR 3.7.2)

passed(q)       = |{students present with marks ≥ target_marks(q)}| ÷ |present|  × 100

success_pct(CO) = Σ_q [ passed(q) × max_marks(q) ] ÷ Σ_q max_marks(q)
```

Marks-weighted, not a flat mean across questions. Your current Python and SQL both use an unweighted mean, which lets a 2-mark question count as much as a 15-mark one.

```
level_CIE(CO) = 3 if success_pct ≥ T3 | 2 if ≥ T2 | 1 if ≥ T1 | else 0
```

`T1/T2/T3` are tenant config with effective-from dates, not constants.

5.2 CO attainment — external (SEE)

```
success_pct_SEE(course) = |{students scoring ≥ target_pct in end-sem}| ÷ |appeared| × 100
level_SEE(CO)           = level(success_pct_SEE)     # identical for every CO of the course
```

Uniform distribution across COs, explicitly documented in the methodology note. This is the accepted practice precisely because AKTU releases only aggregate subject marks.

Do not build the "Dynamic Bayesian Correlation Model" from your v1 plan. Assigning a CO a score derived from a mark no student was measured on is not auditable, and a peer team that asks to see the derivation will not accept it. Keep it as an internal advisory analytics view if you like — it must never feed a SAR table.

5.3 Blend

```
direct(CO)  = w_CIE × level_CIE(CO) + w_SEE × level_SEE(CO)
overall(CO) = w_direct × direct(CO) + w_indirect × indirect(CO)
```

Defaults: `w_CIE=0.30 / w_SEE=0.70` mirroring AKTU's 30:70 marks scheme; `w_direct=0.80 / w_indirect=0.20`. The SAR's own worked example uses 80:20 for SEE:CIE and explicitly notes the weightage is illustrative and programs may choose their own with justification. So: configurable, versioned, and printed in the report.

5.4 CO → PO

```
                Σ_i [ overall(CO_i) × weight(CO_i, PO_j) ]
PO_course(j) = ──────────────────────────────────────────
                        Σ_i weight(CO_i, PO_j)

PO_direct(j)   = mean over all courses addressing PO_j        # NBA-literal default
               | credit-weighted mean                          # opt-in, requires policy note

PO_overall(j)  = w_direct × PO_direct(j) + w_indirect × PO_indirect(j)
```

Output shape must match SAR Tables 3.8.1 / 3.8.2 / 3.8.3 exactly — courses × PO grid, surveys × PO grid, and the three-row summary. Build the export against the real table structure, not an approximation.

5.5 Recomputation

Event-driven, never a "Calculate" button. Emit `marks.ingested`, `mapping.approved`, `survey.closed`, `config.changed` → queue → recompute affected COs → cascade to POs → invalidate caches.

Freeze snapshots per academic year. Once a year is submitted, its computed values are immutable. Editing a config in 2027 must not silently rewrite the 2025 numbers already sitting in a submitted SAR. This is the single most important correctness property in the system and the one that will bite you hardest if you skip it.

5.6 Auto-CQI (SAR Criterion 8, 40 marks)

When `overall(CO) < target`, auto-create a CQI record: gap magnitude, weakest contributing questions, affected cohort, drafted action. Faculty approves or edits. Track closure into the next cycle and auto-write the "action taken and its effect" paragraph.

Criterion 8 is 80 marks and it is where affiliated colleges lose the most, because it requires *two years* of demonstrated closed-loop improvement — 8.1.2 explicitly asks for measures implemented **during two years**. A college that starts collecting this three months before a visit cannot fabricate it. Which means:

your product's value compounds with time-on-platform, and your sales message is "*start now or you cannot claim Criterion 8 in 2028.*" That's the strongest urgency argument available in this market, and it's true.

PART 6 — Architecture

```
Next.js (App Router) + Tailwind + TanStack Table
   │   spreadsheet-feel grid; native Excel paste is a P0 requirement
   ▼
FastAPI (Python 3.12), async, Pydantic v2
   │
   ├── PostgreSQL 16 + RLS + pgvector        core data, attainment SQL, embeddings
   ├── Redis + Celery                        ingestion jobs, survey dispatch, recompute
   ├── Mapping engine (layers 1–8)           pure Python, no I/O, unit-testable
   ├── Parsers                               python-docx (OOXML), pdfplumber, table OCR
   ├── LLM gateway                           provider-abstracted, version-pinned
   └── Object store (S3-compatible)          source documents, immutable
```

Opinions, with reasons:

- **Attainment math lives in SQL.** CTEs over a properly indexed `student_question_marks`. A 120-student course × 8 semesters × 40 courses is a few million rows — trivially fast in Postgres, and the SQL is directly auditable by a technically-literate evaluator. Python orchestrates; it does not compute.
- **pgvector in the same Postgres.** At this scale a separate vector DB is operational overhead with no benefit.
- **Multi-tenancy via Row-Level Security, not application filters.** One forgotten `WHERE institution_id = ?` in an affiliated-college platform is a cross-college marks leak. RLS makes the database enforce it.
- **The mapping engine must be a pure function** with no database or network access, taking a fully-hydrated input struct. That is the only way to write the regression suite you need — and you *need* one, because "the AI changed its mind" is not an acceptable answer when last year's matrix no longer reproduces.
- **Reports via python-docx from templates + LibreOffice headless for PDF.** Templates as versioned config files. NBA revises SAR formats; that must be a template edit, not a release.
- **DPDP Act 2023.** Student marks are personal data. Encryption at rest, hashed survey respondents, documented retention, a signed data-processing agreement per college. Legacy vendors are weak here. Make it a slide in the deck.

RBAC: Faculty → Course Coordinator → HOD → IQAC/NBA Coordinator → Principal → Super Admin. Approval authority for mappings sits at Course Coordinator and above; faculty propose, coordinators endorse. That mirrors how the departmental process actually works and it makes the audit trail meaningful.

PART 7 — Roadmap, team, cost

Assume **4 engineers** (2 backend, 1 frontend, 1 ML/data) + 1 domain person who has actually sat through an NBA visit. That last hire is not optional. Everything expensive in this product is domain knowledge, not code.

Phase 0 — Corpus & gold set (weeks 1–4, parallel to everything) Parse the AKTU syllabus set. Build the ~500-pair labelled CO-PO gold set from published SARs. Encode PO1–PO11 and WK1–WK9 verbatim from the official PDF. This is your asset; start day one.

Phase 1 — Deterministic core (weeks 1–6) Schema, RLS, config versioning, snapshot freezing. SQL attainment engine, both target modes, marks-weighted. Excel/CSV ingest with layout detection. Spreadsheet grid. Exports: CO attainment, Course Articulation Matrix, Program Articulation Matrix. → *Shippable as a free single-faculty tool. No AI in it. Get 100 faculty on this before you build Phase 2.*

Phase 2 — Mapping engine (weeks 5–12) Layers 1–8, WK derivation, CO rewriter, constraint solver, confidence gating, evidence viewer, bulk approve, curriculum gap report, 12→11 migration tool. → *This is the demo that closes departments.*

Phase 3 — Ingestion depth (weeks 11–17) DOCX/PDF QP parser + quality audit. AKTU result ingest. Remembered layouts. Two ERP connectors. Rubric builder for labs/projects/internships.

Phase 4 — Indirect + CQI (weeks 16–21) Four survey types, auto-dispatch, reminders, indirect attainment, CQI generation and closure tracking, SDG mapping.

Phase 5 — SAR assembly (weeks 20–28) Criterion 3 first (most data-driven), then Criterion 1, then Criterion 8, then the outcome-linked parts of Criterion 2. Annexure compilation. Audit-trail export.

Phase 6 — Scale (weeks 28–34) Multi-program, multi-campus, year-on-year comparison, mock-audit checklist against the criteria weightage table, performance work, audit logging.

Budget check. 4 engineers + 1 domain lead for 8 months in India is roughly ₹60–90L fully loaded. Infra under ₹1L/month at pilot scale. LLM spend: one full program's mapping is ~250 courses × 6 COs × 1 batched call ≈ 1,500 calls, i.e. tens of rupees per program per year. **AI cost is a rounding error here** — do not let anyone architect around minimising it. Optimise for correctness and reproducibility instead.

PART 8 — Go to market

Nobody in this market publishes a price. MasterSoft, Creatrix, Kramah, CollPoll — all "request a demo." Creatrix's public-facing model starts around $5,000 per feature with custom engagements reported in the $20k–$500k/year range. That opacity is a six-month sales cycle by design, and it is your opening.

- **Wedge:** free faculty tier. Upload a QP and a marks sheet → CO attainment plus a Course Articulation Matrix in 60 seconds. No demo call, no procurement. Faculty adopt it because the alternative is an evening in Excel.
- **Land:** 8–10 faculty in a department creates demand for the departmental roll-up. First paid seat.
- **Expand:** the NBA coordinator needs Criteria 1, 3 and 8. That's the institutional contract, and it has a hard external deadline attached — sell into the 12 months before a scheduled visit.
- **Price transparently.** ₹75k–₹1.5L per program per year, published. Being the only vendor with a price on the website is a differentiator in a market this opaque, and per-program beats per-student because per-student punishes exactly the large colleges you want.
- **Two urgency levers, both true:** (1) every college still on 12-PO matrices must migrate; (2) Criterion 8 needs two years of closed-loop data, so starting late is unrecoverable.
- **Geographic concentration.** AKTU has one of India's largest affiliated-college networks, concentrated in UP. Twenty colleges in one university makes the next fifty a referral sale. Saharanpur, Meerut, Ghaziabad and Noida are dense with AKTU-affiliated engineering colleges — start where you can drive to the customer.

PART 9 — Risks

Risk	Severity	Mitigation
Peer team rejects AI-generated mappings as not faculty-owned	**High**	Explicit approval logged per row with name/timestamp. Report language: "*faculty-approved, system-assisted.*" Never "*AI-generated.*"
Model produces implausible mapping, gets bulk-approved	**High**	Constraint solver runs after the LLM and rejects violations. Constraints are assertions, not prompts.
Numbers don't reproduce on re-run	**High**	Version-pin model + engine + framework per academic year. Freeze snapshots. Regression suite over the gold set on every deploy.
Cross-tenant data leak	**High**	RLS at the database, not application filters. Pen-test before the first multi-college deployment.
NBA revises SAR again	Medium	Framework versioning is already in the schema. Templates as config. You've now lived through v3→v4; assume v5.
AKTU changes scheme mid-cycle	Medium	Scheme versioning; concurrent schemes per college.
Faculty won't upload	Medium	Excel paste, remembered layouts, ERP sync. The free tool must be genuinely faster than Excel or the whole thesis fails.
DPDP Act exposure	High	Encryption at rest, hashed respondents, retention policy, DPA per college.
Incumbent ERP bundles OBE free	Medium	Compete on what they cannot retrofit: pre-parsed curriculum, WK mapping, evidence-grounded weights, reproducible audit trail.

PART 10 — What to build first, and what not to build

Next four weeks, in order:

1. Parse the AKTU syllabus corpus → structured COs with K-level **arrays**.
2. Encode PO1–PO11 and WK1–WK9 verbatim from the official NBA PDF into a versioned framework table.
3. SQL attainment engine: both target modes, marks-weighted, config-versioned, snapshot-freezing.
4. DOCX question-paper parser reading the CO/KL columns.
5. Wrap it in one free page: upload QP + marks → download CO attainment report and Course Articulation Matrix.

That is a complete, useful, sellable product with **zero AI in it**. The mapping engine is what makes it defensible; the wedge above is what gets you your first hundred users.

Do not build: faculty records, facilities, admissions, fee management, timetabling, attendance, a full LMS, or NAAC/NIRF modules. Criteria 5, 6, 7 and 9 are 440 marks of ERP work that MasterSoft already owns and that adds nothing to your differentiation. Integrate with the ERP; don't become one.

Verify before building report templates: pull the current Tier-II and Tier-I SAR formats fresh from nbaind.org and diff against the January 2025 version this plan is based on. NBA issues notifications between format releases, and building against a stale format is the most expensive rework available in this project.